Porewater sulfate varies spatially and temporally across the coastal terrestrial aquatic interface

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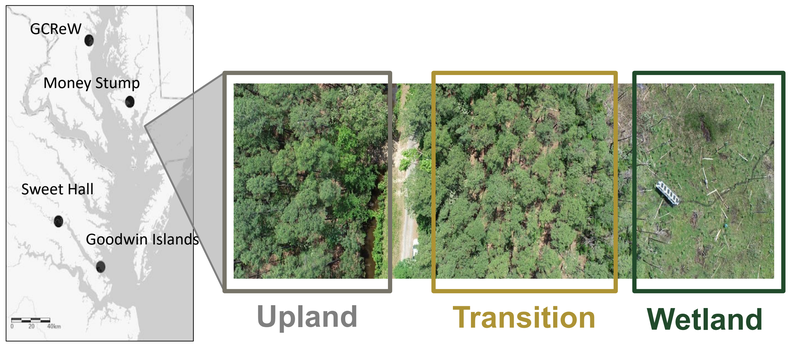
Coastal Observations, Mechanisms, and Predictions Across Systems And Scales
Field, Measurements, And Experiments

Background

- Monitoring transects include upland coastal forest, forest transitioning to wetland, and wetlands in the Chesapeake Bay Region at four sites that are oligohaline, mesohaline, and polyhaline.
- Porewater sampled from 2022-2025 was analyzed for DIC, DOC, Total Alkalinity, CDOM, H_2S, Cl^-, Nutrients, and **SO_4^{2-}**.
- Aquatroll AQ600 sensors are deployed in the field, collecting continuous ground and soil water and salinity values.

Site	Salinity
Sweet Hall	0-2
GCReW	8-10
Money Stump	8-10
Goodwin Islands	18-20



Hypothesis: There is a hierarchical structure to spatial variation in ecosystem structure and function at the coastal terrestrial aquatic interface (TAI) that is set by:

1) the depth and duration of soil saturation

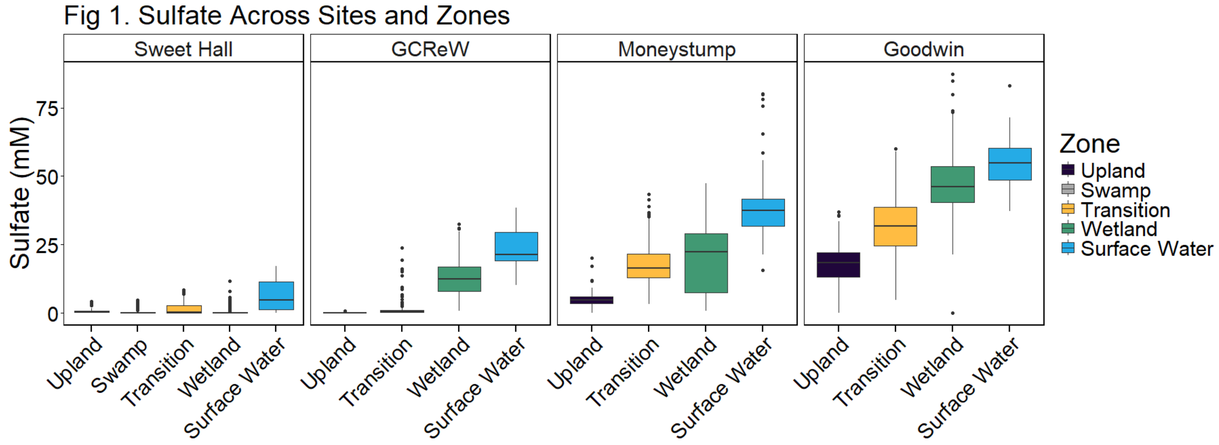
2) the abundance of redox-sensitive elements

3) their interactions with plant traits

Porewater Sulfate

- Porewater was analyzed for Sulfate (SO_4^{2-}, n = 2,293).
- Across sites SO_4^{2-} increased from upland forest through transition zones to the wetland.
- SO_4^{2-} also increased from Sweet Hall site (oligohaline) to Goodwin Islands site (polyhaline).
- Surface water serves as a source of SO_4^{2-} into these ecosystems.

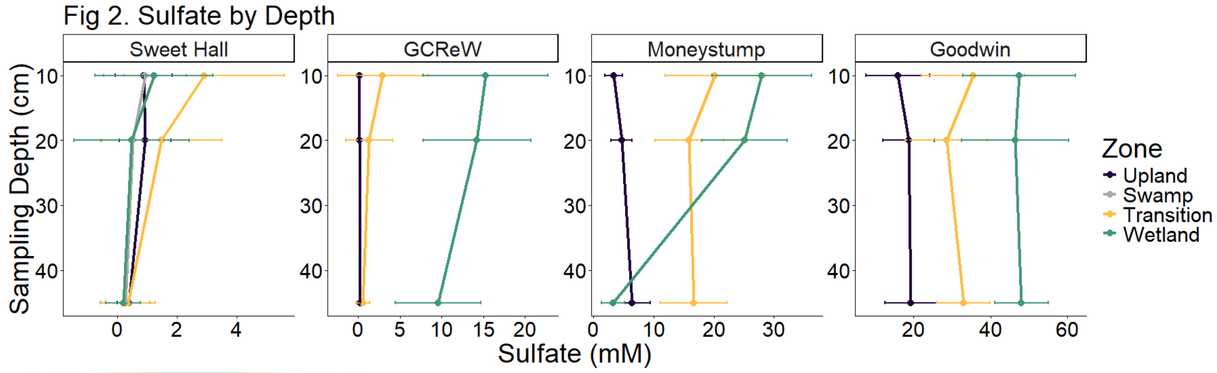
Fig 1. Sulfate Across Sites and Zones



Depth

- Porewater was sampled at three soil depths: 10, 20, and 45 cm.
- SO_4^{2-} concentration changes by depth, but the depth pattern differs across site and zone.

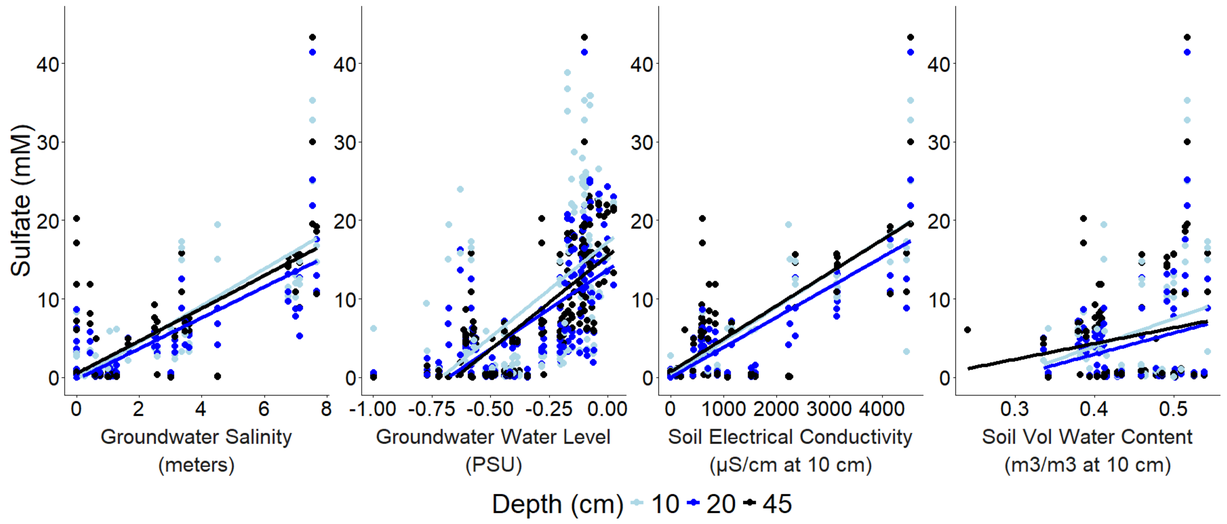
Fig 2. Sulfate by Depth



Environmental variables

- SO_4^{2-} increases with increasing groundwater salinity, groundwater level, soil electrical conductivity, and soil volumetric water content.
- Soil electrical conductivity and soil volumetric water content were measured at 10 cm, but are positively correlated with sulfate at all depths, indicating the 10 cm profile may be influencing or indicative of processes occurring deeper in the soil.

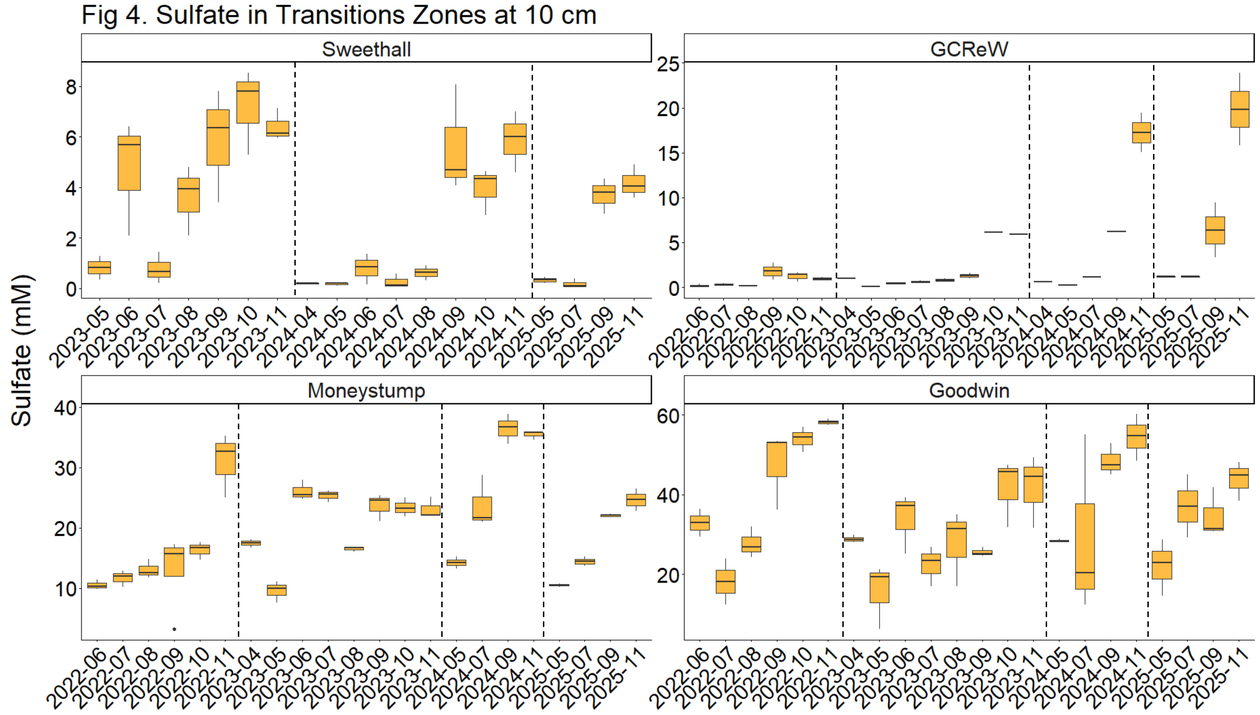
Fig 3. Sulfate versus environmental variables



Seasonal variation in transitioning forests

- SO_4^{2-} tends to increase throughout each year, especially in the fall, and decrease at the start of a new year. This could be because sulfate-reducing microbes are less active as temperatures decrease at the end of each year, or this could be related to changes in precipitation or surface water salinities throughout the year.
- Peak annual SO_4^{2-} appears to be increasing over time at the GCReW Transition site.

Fig 4. Sulfate in Transitions Zones at 10 cm



Take-home messages

- SO_4^{2-}, a redox-sensitive element, varies across depth and season, although this effect differs across different sites in the Chesapeake Bay as well as across zones in the coastal terrestrial aquatic interface.
- SO_4^{2-} was positively correlated with indicators of soil saturation.
- In the future, we can examine the SO_4^{2-} - depth relationship and surface water SO_4^{2-} patterns to determine if there are trends that explain the variability across sites and zones.

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